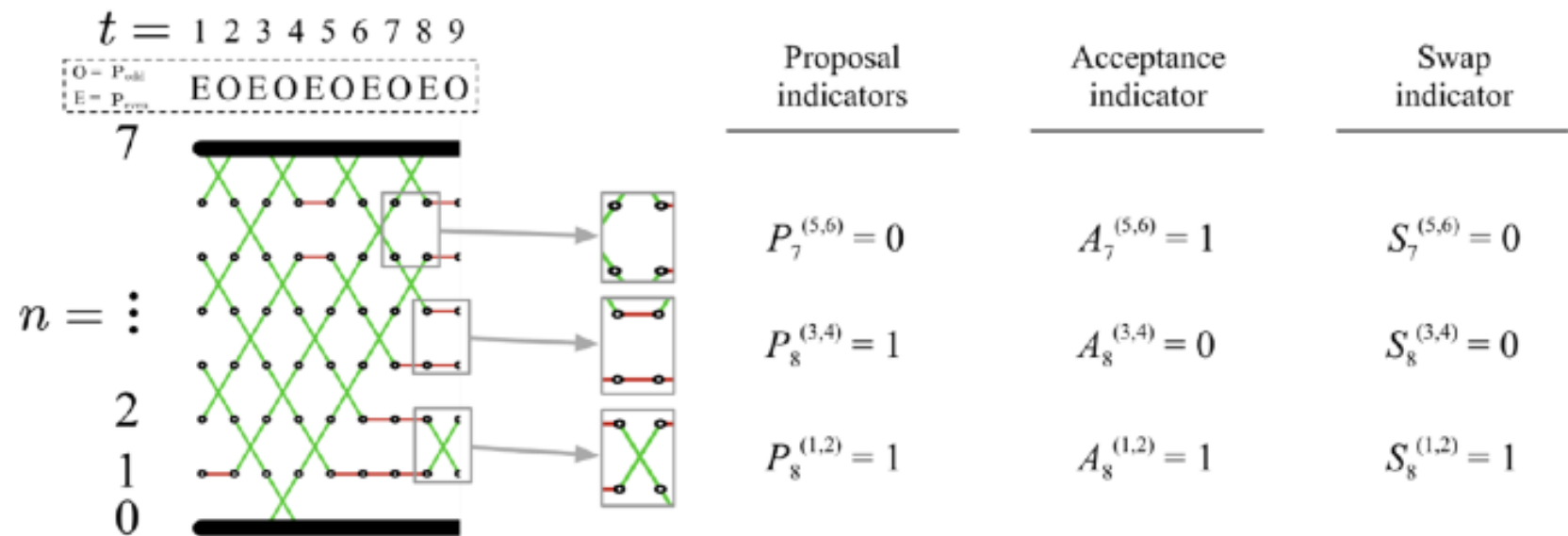




1

8

► Machine m stores:

► State: $Y_t^m \in \mathbb{X}$

► Annealing parameter: $\beta_t^m = \beta_{I_t^m} \in \{\beta_0, \dots, \beta_N\}$

► Proposal direction: $\epsilon_t^m \in \{-1, 1\}$

► Distributed PT state:

$$\mathbf{Y}_t = (Y_t^0, \dots, Y_t^N)$$

$$\mathbf{I}_t = (I_t^0, \dots, I_t^N)$$

$$\boldsymbol{\epsilon}_t = (\epsilon_t^0, \dots, \epsilon_t^N)$$

► We note that $\mathbf{Y}_t$ is $\mathbf{X}_t$ shuffled according to permutation $\mathbf{I}_t$





Dual perspective on PT: swapping parameters not states!



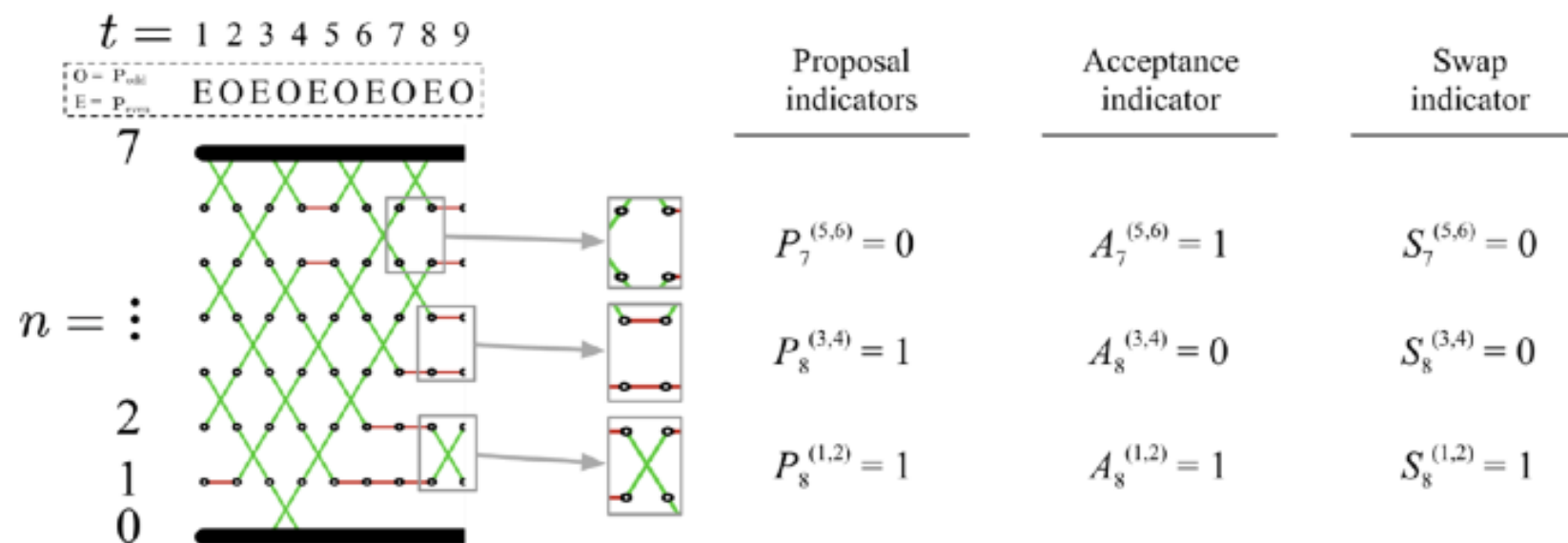


We study communication through the dynamics of the permuting parameters

► X_t^n is commonly referred to as n -th **chain** and Y_t^n the n -th **replica**

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- ▶ **Dual perspective on PT:** Swap annealing parameters not states!
 - ▶ We can study communication through the dynamics of the permuted annealing parameters
 - ▶ More efficient for distributed implementation to swap numbers
- ▶ Machine m stores:
 - ▶ State: $Y_t^m \in \mathbb{X}$
 - ▶ Annealing parameter: $\beta_t^m = \beta_{I_t^m} \in \{\beta_0, \dots, \beta_N\}$
 - ▶ Proposal direction: $\epsilon_t^m \in \{-1, 1\}$
- ▶ Distributed PT state:
 - $\mathbf{Y}_t = (Y_t^0, \dots, Y_t^N)$
 - $\mathbf{I}_t = (I_t^0, \dots, I_t^N)$
 - $\boldsymbol{\epsilon}_t = (\epsilon_t^0, \dots, \epsilon_t^N)$
- ▶ X_t^n is commonly referred to as n -th **chain** and Y_t^m the m -th **replica**
- ▶ We note that $\mathbf{Y}_t$ is $\mathbf{X}_t$ shuffled according to permutation $\mathbf{I}_t$



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